Bitcoin Double Spend Website

Spring 2026

Capstone team project

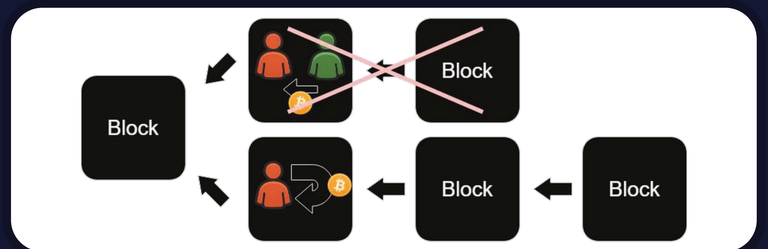
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What is Double Spend?

- Double spending is a threat to the integrity of the Bitcoin blockchain in which an attacker attempts to spend the same Bitcoin in two different transactions.
- The attacker first sends Bitcoin to another user, then creates a second conflicting transaction that sends those same funds back to themselves.
- If the second transaction is confirmed instead of the first, the attacker keeps the Bitcoin, and the original recipient loses the payment and goods.



Summary

The Bitcoin Double Spend Website is a blockchain analysis tool designed to assess and visualize the real-time risk of double spend attacks. By monitoring transactions directly from the peer-to-peer network, the system educates users on the mechanics of blockchain security and transaction finality.

- **Data Acquisition:** Runs a full Bitcoin Core Node on the UAA network to gain direct access to the global mempool and blockchain history.
- **Risk Analysis Engine:** A Python-based service that interfaces with the node to calculate real-time double spend probabilities based on a given transaction's age and number of confirmations on the network.
- **Backend API:** An ASP.NET Core API that orchestrates communication between the analytical engine and the user interface.
- **Frontend & Visualization:** A responsive Blazor web application featuring transaction summaries, interactive graphs to visualize transaction risks, and network trends.



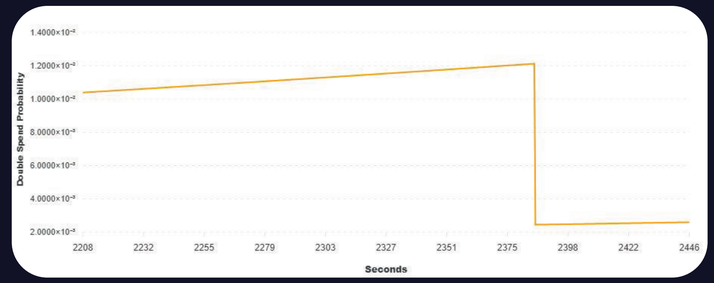
Features

- **Txid Lookup:** A user can search any transaction on the Bitcoin blockchain by its transaction ID (Txid)
- **Detailed Insights:** View transaction age, inputs, outputs, and total BTC spent.
- **Live Risk Metrics:** Assess double spend probability in real-time. Adjust the alpha value to simulate risks from varying attacker hash power.
- **Interactive Visuals:** Track and analyze risk trends over time.

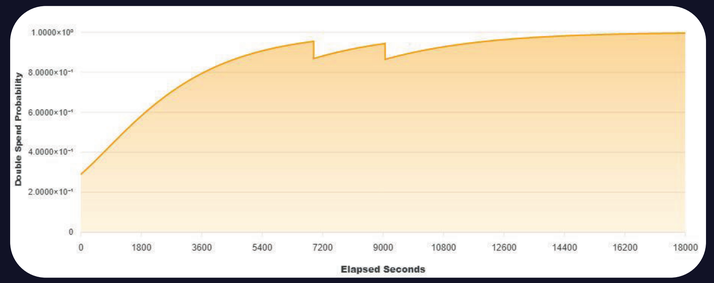
Project Outcomes

- **Education:** Users gain a clearer understanding of what double spend attacks are and what the risks are associated with it.
- **Accessibility:** Traditional blockchain explorers don't commonly provide risk data, limiting how easily users can learn and understand transaction risks. Our platform improves access to these insights through a more user-friendly and informative interface.
- **Usability:** Provides an intuitive interface that is easy to use and conveys information efficiently.

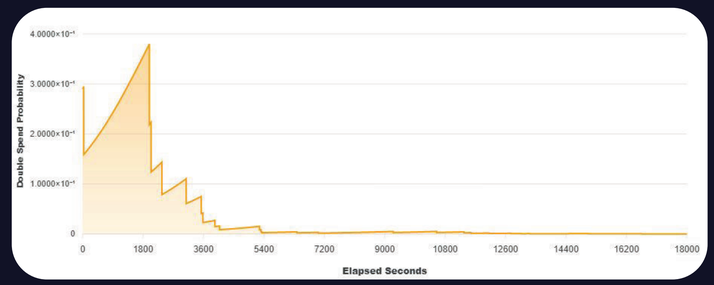
Double Spend Graphs



Real-Time Analysis



5 hr: Abnormal Confirmations



5 hr: Normal Confirmations

References

Bitcoin: A Peer-to-Peer Electronic Cash System (2008).
https://bitcoin.org/bitcoin.pdf
An Analysis of Acceptance Policies for Blockchain Transactions.
IACR ePrint 2018/040 (2018).
https://eprint.iacr.org/2018/040